

ARUNAKIRI VENKATACHALAM

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PROFESSIONAL SUMMARY

Data Scientist with 8+ years of experience building end-to-end AI and data solutions across the automotive and pharmaceutical industries. Specializes in scalable AI tools, ML pipelines, Azure Application Cloud Services and on-premises applications.

TECHNICAL SKILLS

Languages & Frameworks: Python, Django, REST APIs, SQLAlchemy, Streamlit, Linux Shell
Machine Learning & AI: ML Pipelines, NLP, LLM Agents, RAG, MCP, LangChain, Vector Databases
Data Engineering: ETL Pipeline Design, PostgreSQL, Apache Iceberg, AWS S3, High-Frequency Data Processing
DevOps & Cloud: Docker, Kubernetes, CI/CD, Git / GitHub, Keycloak, OAuth 2.0
Analytics & Visualization: Power BI, Grafana, Databricks, Plotly Dash, KPI Dashboard Design

EDUCATION

M.Tech. in Data Science & Engineering · BITS Pilani, Hyderabad — March 2020 – August 2022
Coursework: Machine Learning, Deep Learning, Big Data Analytics, Statistical Modelling, NLP

B.E. in Electronics & Communications · Anna University, Chennai — August 2012 – November 2016
Foundation in signal processing, embedded systems, and communication networks.

PROFESSIONAL EXPERIENCE

Senior Data Scientist · **Volvo Group** — October 2023 – Present

AI & Automation Governance - Framework

- **Business Need:** Decentralized solution development across cross-functional teams led to architecture fragmentation, unvetted AI adoption, and unclear platform hosting strategies.
- **Solution:** Formulating comprehensive enterprise guidelines for the AI & Automation Governance Group within Digital Operations. A review framework to evaluate platform hosting choices and established mandatory architecture, data privacy, and responsible AI usage guidelines.
- **Impact:** Optimizing infrastructure spending, Long-term solution maintainability, standardize application architecture, manage budget allocation and procurement of on-premises infrastructure.

AI Knowledge Assistant for CAB Simulation Engineers

- **Business Need:** CAB simulation engineers spent significant time manually cross-referencing lengthy PDF manuals to find the exact procedures required to run their routine tests.
- **Solution:** Designed and developed a production-ready RAG application enabling engineers to query test requirements, instantly list the required tests, open the source PDFs and navigate directly to the exact page.
- **Impact:** Reduced document search time by approximately 60%, eliminating manual lookups and significantly accelerating the simulation testing lifecycle.
- **Technologies:** Python, LangChain, RAG, LLMs, PostgreSQL, Kubernetes

Hydrogen Engine Data Platform

- **Business Need:** Hydrogen engine test rigs generate high-frequency sensor data requiring automated processing and visualization for engineers.
- **Solution:** Built scalable ETL and ML pipelines to ingest, process, and analyze rig and fleet data; developed interactive Plotly Dash apps and Grafana dashboards; integrated LLM agents with MCP connectors to retrieve live data and generate visualizations within the chat interface.
- **Impact:** Reduced wasted rig hours by 40%, reduced test costs and improved visibility into engine testing operations.
- **Technologies:** Python, Plotly Dash, Grafana, PostgreSQL, Apache Iceberg, AWS S3, LLM Agents, MCP, ML, CI/CD

Data Science Lab Admin Portal

- **Business Need:** User onboarding and resource provisioning for the Data Science Lab relied on manual support requests.
- **Solution:** Developed a full-stack Django app - automated user provisioning, access management, and resource allocation.
- **Impact:** Eliminated manual ticketing and streamlined onboarding for over 200 Data Science Lab users.
- **Technologies:** Django, PostgreSQL, Python, REST APIs, OAuth 2.0, Keycloak

Data Engineering & Analytics Solutions for Volvo

- **Business Need:** Engineering teams required scalable data engineering solutions to process, analyze, and visualize vehicle test and operational data.
- **Solution:** Developed end-to-end data engineering solutions, reusable Python libraries, and standardized data-processing modules supporting multiple analytical programs; conducted structured Python training programs, covering fundamentals and engineering-focused data analysis.
- **Impact:** Improved development efficiency by enabling reusable components and consistent data workflows across projects.
- **Technologies:** Python, SQL, PostgreSQL, ETL Pipelines, Git

MATLAB to Python Migration

- **Business Need:** Existing analytical workflows depended heavily on MATLAB, resulting in high licensing costs and limited flexibility for long-term development.
- **Solution:** Contributed to the MATLAB-to-Python migration initiative, converting analytical applications and establishing Python-based workflows for engineering teams.
- **Impact:** Reduced third-party software licensing costs by 30% while improving maintainability, reproducibility, and adoption of open-source technologies.
- **Technologies:** Python, NumPy, Pandas, SciPy, Git

Machine Learning Solutions for Vehicle Analytics

- **Business Need:** Engineering teams required predictive models to improve vehicle performance analysis and testing.
- **Solution:** Built ML pipelines for tire wear prediction, geo-location tracking, and battery state-of-charge estimation, with automated workflows covering preprocessing, feature engineering, training, validation, and deployment.
- **Impact:** Delivered reusable machine learning solutions that supported engineering analysis and reduced manual effort in model development.
- **Technologies:** Python, Scikit-learn, Pandas, NumPy, Time Series Analysis

Master's Research on Infrastructure Cost Forecasting

- **Business Need:** Insights into infrastructure investment trends were needed to support planning and budget optimization.
- **Solution:** Completed M.Tech thesis analyzing Volvo's on-premises infrastructure costs and developing predictive forecasting models to evaluate future capital expenditure scenarios.
- **Impact:** Identified key cost drivers and provided data-backed recommendations for strategic infrastructure planning.
- **Technologies:** Python, ML, Statistical Modeling, Forecasting, Power BI

Senior Systems Engineer · Infosys — *April 2018 – October 2021*

Document Processing for Unstructured Data

- **Business Need:** Novartis needed to automate extraction of product information from large volumes of unstructured regulatory documents and scanned PDFs to reduce manual processing and improve compliance.
- **Solution:** Designed an OCR-based document processing pipeline to extract structured information from complex multi-format documents, built data validation and error-handling mechanisms, performed root-cause analysis on extraction failures, and designed a PostgreSQL database to centralize extracted product information.
- **Impact:** Achieved 80% extraction accuracy, significantly reduced manual data-entry effort, and improved the reliability and accessibility of regulatory product information.
- **Technologies:** Python, OpenCV, Tesseract, PostgreSQL, ETL Pipelines

Reporting & Operational Monitoring

- **Business Need:** Business and QA teams required visibility into document processing performance, extraction quality, and product information without relying on manual reporting.
- **Solution:** Developed Power BI dashboards to visualize product information and quality metrics and implemented monitoring and logging mechanisms to track processing status, extraction quality, and pipeline failures.
- **Impact:** Replaced manual reporting with automated dashboards and enabled faster identification and resolution of processing issues.
- **Technologies:** Power BI, PostgreSQL, Python, Logging & Monitoring